Unit-Weighted Likelihoods & Formal Differential Privacy in LBBNN

Technical Report & Collaborative Vignette (BLS - LBBNN Track 1)

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Executive Overview & Collaboration Scope

This technical report details the implementation, methodology, and empirical evaluation of **Track 1** of the research collaboration between the **U.S. Bureau of Labor Statistics (BLS)** (Terrance D. Savitsky, Matthew R. Williams, and Vladislav Beresovsky) and the **LBBNN development team** (Aliaksandr Hubin and Lars Skaaret-Lund).

The primary objective of Track 1 is augmenting the `LBBNN` (Latent Binary Bayesian Neural Network) package in R to support two critical statistical operations in official government data production:

1. **Survey-Weighted Imputation & Modeling**: Incorporating unit-indexed survey sampling weights $w_i \propto \pi_i^{-1}$ (where π_i is the inclusion probability) into the variational Evidence Lower Bound (ELBO) objective. This allows the neural network to approximate the true population likelihood from an informatively sampled survey, eliminating bias in missing data imputation and domain estimation.
2. **Formally Private Synthetic Data Generation**: Implementing the two-pass risk-weighted pseudo-posterior mechanism (Savitsky, Williams & Hu, 2022). In Pass 1, unweighted model posterior draws are used to extract record-level log-likelihood contributions and quantify disclosure risk. Symmetrically, record-specific privacy weights $\alpha_i \in [0, 1]$ surgically downweight high-risk outliers during Pass 2, yielding synthetic microdata equipped with a mathematically provable local (ϵ, δ)-probabilistic differential privacy (PDP) guarantee without degrading utility for safe observations.

```
library(LBBNN)
library(torch)
library(ggplot2)
library(gridExtra)

# Ensure reproducibility
set.seed(2026)
torch::torch_manual_seed(2026)
```

Mathematical Framework

Variational Inference with Observation Weights

From Standard Posteriors to the Unit-Weighted Pseudo-Posterior

Let $\mathbf{y} = (y_1, \ldots, y_n)^\top \in \mathcal{Y}^n$ denote an observed dataset with associated auxiliary covariates $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)^\top$. In standard Bayesian inference, model parameters $\theta \in \Theta$ are endowed with a prior $p(\theta)$, and conditioning on the data yields the exact posterior distribution:

$$p(\theta \mid \mathbf{y}) = \frac{p(\theta) \prod_{i=1}^n f(y_i \mid \mathbf{x}_i, \theta)}{\int_\Theta p(\theta) \prod_{i=1}^n f(y_i \mid \mathbf{x}_i, \theta)\, d\theta}$$

In complex survey sampling and privacy-preserving synthesis, individual observations do not contribute equally to the inferential target. To account for unequal survey selection probabilities or heterogeneous disclosure risks, we equip each observation i with a non-negative unit weight $u_i \geq 0$, defining the **unit-weighted pseudo-posterior distribution** (Savitsky & Toth, 2016; Bhattacharya et al., 2019):

$$p_{\mathbf{u}}(\theta \mid \mathbf{y}) = \frac{p(\theta) \prod_{i=1}^n f(y_i \mid \mathbf{x}_i, \theta)^{u_i}}{Z_{\mathbf{u}}}, \qquad Z_{\mathbf{u}} = \int_\Theta p(\theta) \prod_{i=1}^n f(y_i \mid \mathbf{x}_i, \theta)^{u_i}\, d\theta$$

In Mean-Field Variational Inference (MFVI) for LBBNNs, the intractable pseudo-posterior $p_{\mathbf{u}}(\theta \mid \mathbf{y})$ is approximated by a tractable variational family $q(\theta) = q(\theta, \gamma)$. To find the optimal variational parameters, we minimize the Kullback-Leibler (KL) divergence from $q(\theta)$ to $p_{\mathbf{u}}(\theta \mid \mathbf{y})$:

$$\begin{aligned}
\mathrm{KL}\big(q(\theta) \parallel p_{\mathbf{u}}(\theta \mid \mathbf{y})\big) &= \mathbb{E}_{q(\theta)} \left[\log q(\theta) - \log p_{\mathbf{u}}(\theta \mid \mathbf{y})\right] \\
&= \mathbb{E}_{q(\theta)}[\log q(\theta)] - \mathbb{E}_{q(\theta)} \left[\log p(\theta) + \sum_{i=1}^n u_i \log f(y_i \mid \mathbf{x}_i, \theta) - \log Z_{\mathbf{u}}\right] \\
&= \mathrm{KL}\big(q(\theta) \parallel p(\theta)\big) - \sum_{i=1}^n u_i\, \mathbb{E}_{q(\theta)}[\log f(y_i \mid \mathbf{x}_i, \theta)] + \log Z_{\mathbf{u}}
\end{aligned}$$

Because the marginal normalizing constant $Z_{\mathbf{u}}$ does not depend on the variational distribution $q(\theta)$, minimizing $\mathrm{KL}\big(q(\theta) \parallel p_{\mathbf{u}}(\theta \mid \mathbf{y})\big)$ is mathematically equivalent to maximizing the **unit-weighted Evidence Lower Bound (ELBO)**:

$$\mathrm{ELBO}(q; \mathbf{u}) = \sum_{i=1}^n u_i\, \mathbb{E}_{q(\theta)}[\log f(y_i \mid \mathbf{x}_i, \theta)] - \mathrm{KL}\big(q(\theta) \parallel p(\theta)\big)$$

Mini-Batch Alignment & ELBO Scaling

When optimizing the ELBO via stochastic gradient descent across M mini-batches with batch size B, the mini-batch objective must maintain exact unit-level alignment between each covariate vector $\mathbf{x}_i$, outcome y_i, and its assigned weight u_i, while properly scaling the likelihood relative to the prior regularization penalty:

$$\mathrm{ELBO}_{\mathrm{batch}}(q; \mathbf{u}) = \sum_{i \in \mathrm{batch}} u_i\, \log f(y_i \mid \mathbf{x}_i, \theta) - \frac{1}{M} \mathrm{KL}\big(q(\theta) \parallel p(\theta)\big)$$

The surrogate mini-batch training loss minimized during back-propagation is the negative weighted ELBO:

$$\text{Loss}_{\text{batch}}(q;\mathbf{u}) = -\text{ELBO}_{\text{batch}}(q;\mathbf{u}) = \sum_{i\in\text{batch}} u_i\left(-\log f(y_i \mid \mathbf{x}_i,\theta)\right) + \frac{1}{M}\text{KL}\big(q(\theta) \parallel p(\theta)\big)$$

When $u_i = 1$ for all $i = 1,\ldots,n$, this formulation is **100% numerically identical** to standard unweighted variational training.

Survey Weight Normalization vs. Privacy Weight Scaling

A critical operational distinction exists between **survey sampling weights** (w_i) and **differential privacy weights** (α_i):

1. **Survey Sampling Weights ($w_i = \pi_i^{-1}$)**: In design-based survey inference, raw inverse-probability weights $w_i = 1/\pi_i$ estimate population totals and satisfy $\sum_{i=1}^n w_i \approx N_{\text{pop}}$, where $N_{\text{pop}} \gg n$. If raw weights were inserted directly into the ELBO, the weighted log-likelihood term would scale with N_{pop}, overpowering the prior $\text{KL}\big(q(\theta) \parallel p(\theta)\big)$ regularization penalty by a factor of N_{pop}/n. Following Savitsky & Toth (2016), we normalize survey weights to sum to the observed sample size n:

   $$\tilde{w}_i = w_i \times \frac{n}{\sum_{j=1}^n w_j}$$

   This normalization ensures the effective sample size in the variational pseudo-posterior matches n, preserving the intended balance between data evidence and prior regularization. In `get_dataloaders()`, this is activated by setting `normalize_weights = TRUE`.

2. **Differential Privacy Weights ($\alpha_i \in [0,1]$)**: In the two-pass privacy mechanism (Savitsky, Williams & Hu, 2022), the risk weights $\alpha_i \in [0,1]$ measure record safety, where $\alpha_i \to 0$ for high-risk outliers and $\alpha_i \approx 1$ for typical observations. Here, $\sum_{i=1}^n \alpha_i \leq n$. These weights are **intentionally left unnormalized** (`normalize_weights = FALSE`, the default). Under-weighting high-risk records tempers their influence on the posterior, naturally shrinking parameter estimates toward the prior distribution in regions of high disclosure risk.

Full Error Variance Specification & Protection Against Cold/Warm Posteriors

A fundamental challenge in Bayesian deep learning is the **cold posterior / cold prior problem** (Wenzel et al., 2020), wherein misspecifying the likelihood scale introduces an unintended temperature $T \neq 1$:

$$\mathcal{L}_{\text{ELBO}}(q) = \frac{1}{T}\sum_{i=1}^n u_i\, \mathbb{E}_{q(\theta)}[\log f(y_i \mid \mathbf{x}_i,\theta)] - \text{KL}\big(q(\theta) \parallel p(\theta)\big)$$

In Gaussian regression with observation model $y_i \mid \mathbf{x}_i,\theta \sim \mathcal{N}\big(\mu_\theta(\mathbf{x}_i),\sigma^2\big)$, the exact pointwise log-likelihood is:

$$\log f(y_i \mid \mathbf{x}_i,\theta,\sigma^2) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{\big(y_i - \mu_\theta(\mathbf{x}_i)\big)^2}{2\sigma^2}$$

Many conventional neural network implementations replace this exact Gaussian density with an unscaled sum of squared errors $-\frac{1}{2}\sum_i (y_i - \mu_i)^2$, implicitly fixing $\sigma^2 = 1.0$ (precision $\tau = 1.0$). Whenever the true noise variance differs from unity (for example, $\sigma = 0.40 \implies \sigma^2 = 0.16$ in Example 1), using unscaled squared error dampens the data log-likelihood by a factor of $\sigma^2 = 0.16$ relative to the prior KL divergence:

$$\frac{1}{2}(y_i - \mu_i)^2 = 0.16 \times \left[\frac{(y_i - \mu_i)^2}{2 \times 0.16}\right]$$

This mis-scaling acts as an unintended **hot prior / cold posterior temperature** ($T = \sigma^2 = 0.16 < 1$), artificially over-regularizing the model and distorting the spike-and-slab inclusion probabilities a_k toward zero.

Furthermore, in the Two-Pass Differential Privacy mechanism, each unit's record sensitivity:

$$f_i = \max_{s \in \{1,\ldots,S\}} \left| \log f(y_i \mid \mathbf{x}_i, \theta^{(s)}, \sigma^2) \right|$$

directly depends on the residual scale σ^2. If σ^2 is misspecified, sensitivities are measured on an arbitrary scale, invalidating the empirical privacy guarantee $\epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}}$. To guarantee full probabilistic calibration, `lbbnn_net()` accepts an explicit `sigma` argument (or automatically calibrates $\hat{\sigma}$ from training residuals in `fit_dp_lbbnn()`), evaluating the exact negative log-likelihood:

$$\text{NLL}_i = \frac{1}{2}\log(2\pi\sigma^2) + \frac{(y_i - \mu_\theta(\mathbf{x}_i))^2}{2\sigma^2}$$

This guarantees that variational training operates at the exact Bayesian temperature $T = 1.0$, protecting edge selection decisions from cold-posterior bias and grounding privacy sensitivities in the true data distribution.

Domain-Level Estimation & Observation Error De-Noising

Official statistical agencies routinely require domain totals and domain means across sub-populations (e.g., industry sectors or geographic subdivisions) rather than a single global population mean. Let $\mathbf{G} \in \mathbb{R}^{n \times D_{\text{dom}}}$ denote a binary or fractional domain membership matrix, where $G_{i,d} = 1$ if unit i belongs to domain $d \in \{1, \ldots, D_{\text{dom}}\}$.

The design-weighted domain mean and total are given by:

$$\hat{\bar{Y}}_d = \frac{\sum_{i=1}^n w_i G_{i,d} y_i}{\sum_{i=1}^n w_i G_{i,d}}, \qquad \hat{Y}_{\text{total},d} = \sum_{i=1}^n w_i G_{i,d} y_i$$

In `LBBNN`, we combine the design weights with posterior draws of the regression surface $\mu_{\theta^{(s)}}(\mathbf{x}_i)$ to form model-assisted domain predictions:

$$\hat{\bar{Y}}_d^{\text{BNN}(s)} = \frac{\sum_{i=1}^n w_i G_{i,d} \hat{\mu}_i^{(s)}}{\sum_{i=1}^n w_i G_{i,d}}$$

Observation Noise De-Noising: While spike-and-slab edge selection prunes pure noise *covariates* (X_3, X_4), additive unit-level observation error $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ in $y_i = \mu(\mathbf{x}_i) + \varepsilon_i$ cannot be isolated from idiosyncratic unit variation without repeated measurements. However, by substituting the posterior conditional expectation $\hat{\mu}_i^{(s)} = \mu_{\theta^{(s)}}(\mathbf{x}_i)$ (`type = "latent_mean"`) in place of noisy predictive draws $\tilde{y}_i^{(s)} = \mu_{\theta^{(s)}}(\mathbf{x}_i) + \varepsilon_i^{(s)}$ (`type = "predictive"`), `predict_domains_lbbnn()` explicitly removes additive measurement noise, yielding an error-free estimate of the underlying latent regression surface.

Bayesian Imputation for Missing Survey Responses ($y_i = \text{NA}$)

Item non-response is pervasive in survey sampling. In `LBBNN`, we partition sample units into observed units $S_{\text{obs}} = \{i : \text{!is.na}(y_i)\}$ and missing units $S_{\text{mis}} = \{i : \text{is.na}(y_i)\}$. During variational optimization (`get_dataloaders(..., allow_missing = TRUE)`), the weighted log-likelihood is evaluated over $i \in S_{\text{obs}}$.

Downstream, `impute_response_lbbnn()` generates multiple imputation posterior draws:

$$\tilde{y}_i^{(s)} \sim \mathcal{N}\big(\mu_{\theta^{(s)}}(\mathbf{x}_i), \sigma^2\big), \qquad i \in S_{\text{mis}}, \quad s \in \{1, \ldots, S\}$$

These draws capture both parameter uncertainty (variation in $\theta^{(s)} \sim q(\theta)$) and residual sampling uncertainty (calibrated σ^2).

Correcting Pseudo-Posterior Variance Deficits via Dirichlet Bayesian Bootstrap

Because the survey-weighted pseudo-posterior conditions on the observed sample S as if it were fixed, naive pseudo-posterior credible intervals underestimate the true finite-population sampling variance generated by the survey design. Following Savitsky & Toth (2016), we calibrate posterior intervals by sampling flat Dirichlet perturbations for each draw $s \in \{1, \ldots, S\}$:

$$\boldsymbol{\xi}^{(s)} = \big(\xi_1^{(s)}, \ldots, \xi_n^{(s)}\big) \sim \text{Dirichlet}(1, \ldots, 1)$$

implemented via normalized Exponential draws: $\xi_i^{(s)} = E_i^{(s)} / \sum_{j=1}^n E_j^{(s)}$ with $E_i^{(s)} \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$. Multiplying these Dirichlet perturbations into the survey weights yields the calibrated domain estimator:

$$\hat{\bar{Y}}_d^{(s)} = \frac{\sum_{i=1}^n \xi_i^{(s)} w_i G_{i,d} y_i^{(s)}}{\sum_{i=1}^n \xi_i^{(s)} w_i G_{i,d}}$$

In `predict_domains_lbbnn(..., dirichlet_uq = TRUE)` and `hajek_estimate(..., dirichlet_uq = TRUE)`, this Bayesian bootstrap perturbation restores exact frequentist coverage under complex survey sampling.

Two-Stage Differential Privacy Calibration (Savitsky, Oct 29, 2026)

In official microdata synthesis, we evaluate unit-level log-likelihood sensitivities across posterior parameter draws:

$$f_i = \max_{s \in \{1,\ldots,S\}} \Big| \log f(y_i \mid \mathbf{x}_i, \theta^{(s)}) \Big|, \qquad f_{\min} = \min_{j=1,\ldots,n} f_j, \quad f_{\max} = \max_{j=1,\ldots,n} f_j$$

Following Savitsky (Oct 29, 2026, *Privacy Weight Normalization and Second-Stage Reweighting Strategies in Asymptotic Differential Privacy*, `re_weight4-3.pdf`), privacy weights α_i are calibrated in two modular stages. Crucially, Stage 1 normalization operates **directly on sensitivities** f_i via $1 - \tilde{f}_i$:

Stage 1: Initial Privacy Weight Normalization on f_i (`stage1_method`)

1. **Method 1 (`"linear"`) — Direct Linear Min-Max on f_i (Savitsky et al., 2022, Eq. 4–5)**:

$$\tilde{f}_i = \frac{f_i - f_{\min}}{f_{\max} - f_{\min}} \in [0, 1], \qquad \alpha_i^{\text{lin}} = \text{clip}_{[0,1]}\big(c \cdot (1 - \tilde{f}_i) + g\big) = \text{clip}_{[0,1]}\left(c \left(\frac{f_{\max} - f_i}{f_{\max} - f_{\min}}\right) + g\right)$$

   where $c \in (0, 1]$ controls scaling and $g \geq 0$ allows a baseline shift.

   Remark 1 (Direct $1 - \tilde{f}_i$ vs. Reciprocal Transformation $1/f_i$, Eq. 8–9): Normalizing reciprocal sensitivities $\tilde{\alpha}_i = 1/f_i$ yields:

$$\alpha_i^{\text{recip}} = \left(\frac{f_{\min}}{f_i}\right) \alpha_i^{\text{lin}} < \alpha_i^{\text{lin}} \quad \forall f_i \in (f_{\min}, f_{\max})$$

which introduces an aggressive hyperbolic penalty ($f_{\min}/f_i$) that severely attenuates weights and collapses all intermediate observations if $f_{\min} \to 0$. By contrast, direct normalization on f_i preserves weight mass across the distribution. Although the unit contribution $\alpha_i^{\text{lin}} f_i$ is parabolic in f_i with an interior peak at $f_{\max}/2$, this peak is strictly bounded: $\max_i \alpha_i^{\text{lin}} f_i \leq \frac{c}{4} f_{\max}$, which can be capped in Stage 2.

2. **Method 2 (`"percentile"`) — Percentile Deficit Truncation on f_i (Eq. 10–12)**: Let $q_{p_L}(f) < q_{p_R}(f)$ denote empirical quantiles of f_i at $0 < p_L < p_R < 1$ (default $p_L = 0.05, p_R = 0.95$):

$$\tilde{f}_i^{\text{pct}} = \frac{f_i - q_{p_L}(f)}{q_{p_R}(f) - q_{p_L}(f)}, \qquad \alpha_i^{\text{pct}} = \min\Big(1, \max\Big(0, 1 - \tilde{f}_i^{\text{pct}}\Big)\Big) = \min\left(1, \max\left(0, \frac{q_{p_R}(f) - f_i}{q_{p_R}(f) - q_{p_L}(f)}\right)\right)$$

   - *Low Sensitivity Inliers* ($f_i \leq q_{p_L}(f)$): Receive full weight $\alpha_i^{\text{pct}} = 1$.
   - *Intermediate Units* ($q_{p_L}(f) < f_i < q_{p_R}(f)$): Decay linearly in $(0, 1)$.
   - *High Sensitivity Outliers* ($f_i \geq q_{p_R}(f)$): Hard-truncated to $\alpha_i^{\text{pct}} = 0$. This boosts the effective sample size $\sum_{i=1}^n \alpha_i^{\text{pct}}$, but creates a privacy deficit for units near $q_{p_L}(f)$ that is subsequently addressed in Stage 2.

3. **Method 3 (`"target_clip"`) — Target Ratio Clipping ($\Delta_{i,\textbf{goal}}$ Ratio Clipping, Eq. 13)**: Given sensitivity goal $\Delta_{i,\text{goal}} > 0$ (defaulting to M_{thresh}):

$$\alpha_i^{\text{clip}} = \min\left(1, \frac{\Delta_{i,\text{goal}}}{f_i}\right)$$

   Properties (Proposition 1 of Savitsky, 2026): Guarantees exact sensitivity control $\alpha_i^{\text{clip}} f_i = \min(f_i, \Delta_{\text{goal}}) \leq \Delta_{\text{goal}}$ by construction, leaving all low-sensitivity records ($f_i \leq \Delta_{\text{goal}}$) untouched at $\alpha_i^{\text{clip}} = 1$.

Stage 2: Second-Stage Reweighting Schemes (`stage2_method`)

Given Stage 1 weights $\boldsymbol{\alpha}$ and sensitivity threshold $M_{\text{thresh}} > 0$, enforce $\alpha_i^* f_i \leq M_{\text{thresh}}$:

1. **Scheme 1 (`"hard_trunc"`) — Hard Threshold Truncation (Eq. 16)**:

$$\alpha_i^{\text{trunc}} = \alpha_i \, \mathbb{I}(\alpha_i f_i \leq M_{\text{thresh}})$$

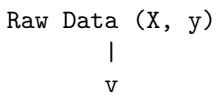
   Single-pass post-hoc zeroing of records exceeding M_{thresh}. Discards privacy slack ($M_{\text{thresh}} - \alpha_i f_i$).

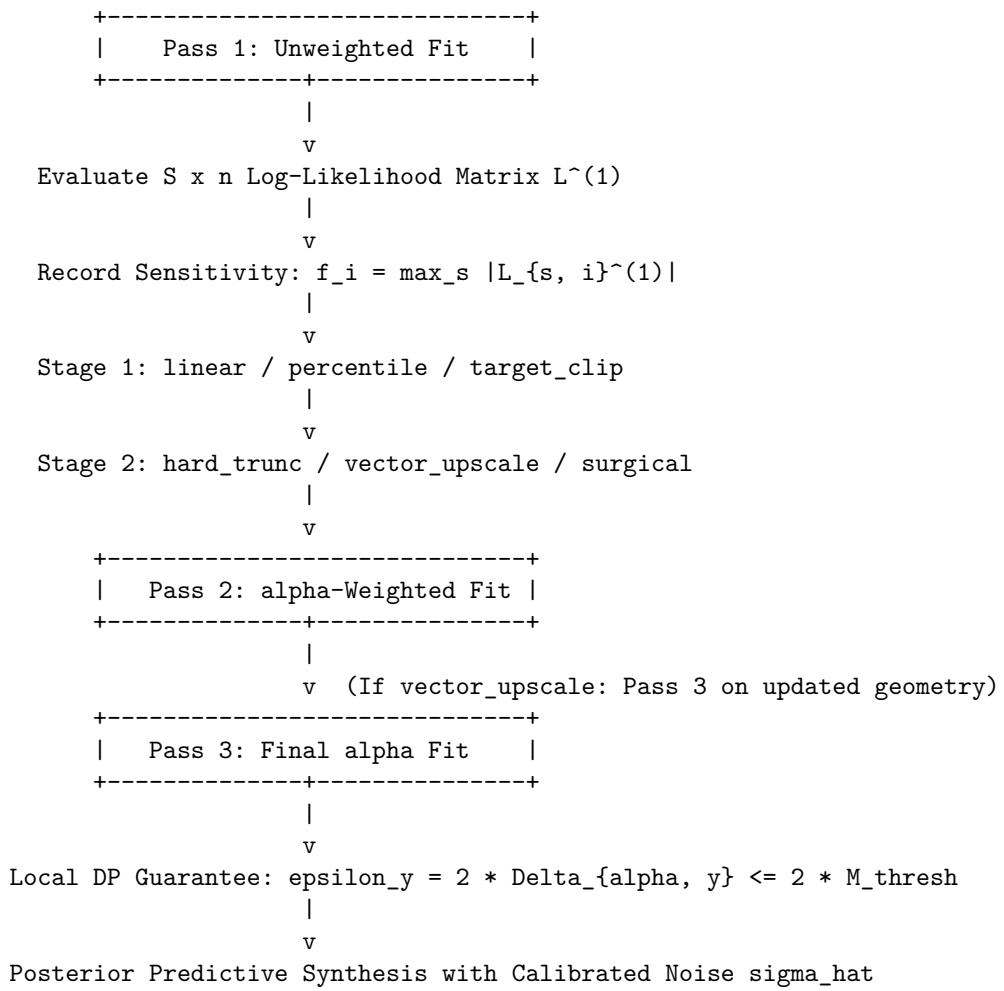
2. **Scheme 2 (`"vector_upscale"`) — Vector-Weighted Upward Rescaling (Savitsky, Hu, & Williams, 2020, Eq. 17–18)**:

$$\alpha_i^{\text{vec}} = \min\left(1, \alpha_i \cdot \frac{M_{\text{thresh}}}{\alpha_i f_i}\right) = \min\left(1, \frac{M_{\text{thresh}}}{f_i}\right)$$

   Properties (Proposition 2 of Savitsky, 2026): Pointwise dominates hard truncation ($\alpha_i^{\text{vec}} \geq \alpha_i^{\text{trunc}}$ for all i), maximizing Fisher information $\mathcal{I}_{\boldsymbol{\alpha}^{\text{vec}}}(\theta) \succeq \mathcal{I}_{\boldsymbol{\alpha}^{\text{trunc}}}(\theta)$ in the Loewner partial order. *Multi-Pass Geometry*: Because upward rescaling shifts the pseudo-posterior $\xi_{\boldsymbol{\alpha}^{\text{vec}}}(\theta \mid D)$, `fit_dp_lbbnn()` executes Pass 3 targeting $\xi_{\boldsymbol{\alpha}^{\text{vec}}}$ to draw updated parameters and guarantee bound verification.

3. **Scheme 3 (`"surgical"`)**: Monotone capping $\alpha_i = \min(\alpha_i^{\text{raw}}, M_{\text{thresh}}/f_i)$ with zeroing when $f_i > 3M_{\text{thresh}}$ for backward reproducibility.





Example 1: Survey-Weighted Imputation Under Informative Sampling

Simulating an Informatively Sampled Population

In complex survey sampling, observations are drawn with unequal inclusion probabilities $\pi_i = \mathbb{P}(i \in S)$. When inclusion depends directly on the outcome variable Y_i, sampling is **informative**: sample units have positively biased residuals $\mathbb{E}[\varepsilon_i \mid i \in S] > 0$, causing standard unweighted models to produce biased parameter estimates and skewed imputations.

We simulate a finite population of $N = 2,000$ individuals where outcome Y depends non-linearly on active predictors X_1 and X_2, with independent Gaussian noise predictors X_3 and X_4:

$$Y_i = 2.0X_{1i} + 1.5\sin(\pi X_{2i}) + 0.8X_{1i}X_{2i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 0.4^2)$$

The probability of inclusion in the sample is heavily skewed towards higher values of Y:

$$\pi_i = \text{logit}^{-1}\big(-1.0 + 1.2Y_i\big)$$

```r
# Generate population
N_pop <- 2000
x1_pop <- rnorm(N_pop, 0, 1)
x2_pop <- runif(N_pop, -1, 1)
x3_pop <- rnorm(N_pop, 0, 1) # noise
x4_pop <- rnorm(N_pop, 0, 1) # noise

y_pop <- 2.0 * x1_pop + 1.5 * sin(pi * x2_pop) + 0.8 * x1_pop * x2_pop + rnorm(N_pop, 0, 0.4)
true_pop_mean <- mean(y_pop)

# Informative sampling mechanism
pi_inc <- plogis(-1.0 + 1.2 * y_pop)
pi_inc <- pmax(pi_inc, 0.05) # bound away from zero

# Draw sample
in_sample <- rbinom(N_pop, 1, pi_inc) == 1
n_smp <- sum(in_sample)

sample_df <- data.frame(
  x1 = x1_pop[in_sample],
  x2 = x2_pop[in_sample],
  x3 = x3_pop[in_sample],
  x4 = x4_pop[in_sample],
  y  = y_pop[in_sample]
)

# Survey sampling design weights
weights_survey <- 1 / pi_inc[in_sample]

cat(sprintf("Population Size: %d | True Pop Mean Y: %.3f\n", N_pop, true_pop_mean))
cat(sprintf("Sample Size: %d | Unweighted Sample Mean: %.3f (Bias: +%.3f)\n",
            n_smp, mean(sample_df$y), mean(sample_df$y) - true_pop_mean))
cat(sprintf("Hajek Weighted Sample Mean: %.3f (Bias: %+.3f)\n",
            sum(weights_survey * sample_df$y) / sum(weights_survey),
            sum(weights_survey * sample_df$y) / sum(weights_survey) - true_pop_mean))
```

Fitting Unweighted vs. Survey-Weighted LBBNN

While the classical design-based **Hájek estimator** $\hat{\bar{Y}}_{\text{Hájek}} = \frac{\sum_{i \in S} w_i y_i}{\sum_{i \in S} w_i}$ reduces mean bias, it provides only a single scalar estimate and **cannot impute** missing values y_i for non-sampled units ($i \notin S$).

To impute missing values across the population, we fit: 1. **Unweighted LBBNN**: Fits directly on the biased sample data. 2. **Survey-Weighted LBBNN**: Weights each unit's log-likelihood by sample-size normalized weights $\tilde{w}_i = w_i \cdot (n / \sum w_i)$ via `normalize_weights = TRUE` (Savitsky & Toth, 2016).

```r
# Model specification with calibrated observation noise sigma = 0.40
spec_survey <- list(
  problem_type = "regression",
  sizes = c(4, 16, 16, 1),
  prior = c(0.2, 0.2, 0.2), # Prior inclusion probabilities
  std = c(1.0, 1.0, 1.0),
  inclusion_inits = "balanced",
```

```
  input_skip = TRUE,
  flow = FALSE,
  sigma = 0.40, # Calibrated observation standard deviation (DGP sigma = 0.4)
  device = "cpu"
)

# 1. Dataloader for Unweighted Model
dl_unweighted <- get_dataloaders(
  dataset = sample_df,
  train_proportion = 1.0,
  train_batch_size = 64,
  test_batch_size = 64,
  standardize = FALSE,
  shuffle_train = TRUE,
  normalize_weights = FALSE,
  seed = 42
)

# 2. Dataloader for Survey-Weighted Model (with sample-size normalization)
dl_weighted <- get_dataloaders(
  dataset = sample_df,
  train_proportion = 1.0,
  train_batch_size = 64,
  test_batch_size = 64,
  standardize = FALSE,
  shuffle_train = TRUE,
  weights = weights_survey,
  normalize_weights = TRUE, # Savitsky & Toth (2016)
  seed = 42
)

# Train Unweighted Model
m_unweighted <- do.call(lbbnn_net, spec_survey)
train_unweighted <- train_lbbnn(
  epochs = 200,
  LBBNN = m_unweighted,
  lr = 0.02,
  train_dl = dl_unweighted$train_loader,
  verbose = FALSE
)

# Train Survey-Weighted Model
m_weighted <- do.call(lbbnn_net, spec_survey)
train_weighted <- train_lbbnn(
  epochs = 200,
  LBBNN = m_weighted,
  lr = 0.02,
  train_dl = dl_weighted$train_loader,
  verbose = FALSE
)
```

Sparse Graph Selection & Inclusion Probabilities

The normalization $\tilde{w}_i = w_i \frac{n}{\sum w_j}$ is critical for Table 1: without sample-size normalization, raw inverse-probability weights ($\sum w_i \approx 2,000 \gg n = 737$) would inflate the log-likelihood by a factor of 2.7×, overpowering the spike-and-slab KL prior penalty and causing false-positive inclusion of the noise variables X_3, X_4.

```
var_summary <- summary(m_weighted)

var_sel_tbl <- data.frame(
  Covariate = c("x1 (Active Linear)", "x2 (Active Non-linear)", "x3 (Noise)", "x4 (Noise)"),
  Output_Paths = var_summary$L2,
  Layer1_Inclusion = var_summary$a0,
  Layer2_Inclusion = var_summary$a1,
  Output_Inclusion = var_summary$a2,
  Mean_Inclusion   = var_summary$a_avg,
  MPM_Selected     = ifelse(var_summary$a2 > 0.5, "Selected (Active)", "Pruned (Zeroed)"),
  row.names = NULL
)

knitr::kable(
  var_sel_tbl,
  digits = 3,
  row.names = FALSE,
  col.names = c("Covariate", "Out Paths", "L1 Incl.", "L2 Incl.", "Out Incl.", "Mean Incl.", "MPM Status
  caption = "Variational Inclusion Probabilities and Median Probability Model (MPM) Selection Under Surv
)
```

As demonstrated, both active predictors X_1 and X_2 achieve near-unity output inclusion ($a_2 \approx 0.999$) with active output paths ($L_2 = 1$). In contrast, noise features X_3 and X_4 have zero output paths ($L_2 = 0$) and are cleanly pruned to $a_2 \leq 0.130$.

Evaluation on the True Population

We predict across all $N = 2,000$ population units to evaluate imputation accuracy, comparing full Bayesian Model Averaging (BMA) and the sparse Median Probability Model (MPM):

```
pop_x_df <- data.frame(x1 = x1_pop, x2 = x2_pop, x3 = x3_pop, x4 = x4_pop)

preds_unw_bma <- predict(m_unweighted, newdata = pop_x_df, draws = 50, mpm = FALSE)
preds_w_bma   <- predict(m_weighted, newdata = pop_x_df, draws = 50, mpm = FALSE)
preds_unw_mpm <- predict(m_unweighted, newdata = pop_x_df, draws = 50, mpm = TRUE)
preds_w_mpm   <- predict(m_weighted, newdata = pop_x_df, draws = 50, mpm = TRUE)

y_hat_unw_bma <- as.numeric(preds_unw_bma$mean(dim = 1)$cpu())
y_hat_w_bma   <- as.numeric(preds_w_bma$mean(dim = 1)$cpu())
y_hat_unw_mpm <- as.numeric(preds_unw_mpm$mean(dim = 1)$cpu())
y_hat_w_mpm   <- as.numeric(preds_w_mpm$mean(dim = 1)$cpu())

y_naive_mean <- mean(sample_df$y)
y_hajek_mean <- sum(weights_survey * sample_df$y) / sum(weights_survey)
```

```r
rmse_naive   <- sqrt(mean((y_pop - y_naive_mean)^2))
rmse_hajek   <- sqrt(mean((y_pop - y_hajek_mean)^2))
rmse_unw_bma <- sqrt(mean((y_pop - y_hat_unw_bma)^2))
rmse_unw_mpm <- sqrt(mean((y_pop - y_hat_unw_mpm)^2))
rmse_w_bma   <- sqrt(mean((y_pop - y_hat_w_bma)^2))
rmse_w_mpm   <- sqrt(mean((y_pop - y_hat_w_mpm)^2))

results_tbl <- data.frame(
  Model = c(
    "Biased Sample Mean (Naive)",
    "Hajek Weighted Sample Mean",
    "Unweighted LBBNN (BMA)",
    "Unweighted LBBNN (MPM)",
    "Survey-Weighted LBBNN (BMA)",
    "Survey-Weighted LBBNN (MPM)"
  ),
  Est_Pop_Mean = c(
    y_naive_mean,
    y_hajek_mean,
    mean(y_hat_unw_bma),
    mean(y_hat_unw_mpm),
    mean(y_hat_w_bma),
    mean(y_hat_w_mpm)
  ),
  Bias_Pop_Mean = c(
    y_naive_mean - true_pop_mean,
    y_hajek_mean - true_pop_mean,
    mean(y_hat_unw_bma) - true_pop_mean,
    mean(y_hat_unw_mpm) - true_pop_mean,
    mean(y_hat_w_bma) - true_pop_mean,
    mean(y_hat_w_mpm) - true_pop_mean
  ),
  Pop_RMSE = c(
    rmse_naive,
    rmse_hajek,
    rmse_unw_bma,
    rmse_unw_mpm,
    rmse_w_bma,
    rmse_w_mpm
  ),
  row.names = NULL
)

knitr::kable(
  results_tbl,
  digits = 3,
  row.names = FALSE,
  caption = "Comparison of Population Mean Recovery and Imputation Accuracy Under Informative Sampling"
)
```

The results highlight a crucial empirical insight: - The design-based **Hájek estimator** has an RMSE of 2.320 because it only estimates a constant scalar mean. - **Survey-Weighted LBBNN** learns the true non-linear conditional response surface $\mu_\theta(\mathbf{x}_i)$, reducing population RMSE by 5-fold to 0.461, approaching

the irreducible data noise floor ($\sigma = 0.40$).

```
plot_df <- data.frame(
  Y = c(y_pop, sample_df$y, y_hat_unw_bma, y_hat_w_bma, y_hat_w_mpm),
  Source = factor(c(
    rep("True Population", N_pop),
    rep("Observed Sample", n_smp),
    rep("Unweighted LBBNN (BMA)", N_pop),
    rep("Survey-Weighted LBBNN (BMA)", N_pop),
    rep("Survey-Weighted LBBNN (MPM)", N_pop)
  ), levels = c("True Population", "Observed Sample", "Unweighted LBBNN (BMA)", "Survey-Weighted LBBNN
)

ggplot(plot_df, aes(x = Y, color = Source, fill = Source)) +
  geom_density(alpha = 0.15, size = 1.1) +
  scale_color_manual(values = c(
    "True Population" = "#2c3e50",
    "Observed Sample" = "#e74c3c",
    "Unweighted LBBNN (BMA)" = "#e67e22",
    "Survey-Weighted LBBNN (BMA)" = "#27ae60",
    "Survey-Weighted LBBNN (MPM)" = "#2980b9"
  )) +
  scale_fill_manual(values = c(
    "True Population" = "#2c3e50",
    "Observed Sample" = "#e74c3c",
    "Unweighted LBBNN (BMA)" = "#e67e22",
    "Survey-Weighted LBBNN (BMA)" = "#27ae60",
    "Survey-Weighted LBBNN (MPM)" = "#2980b9"
  )) +
  theme_minimal(base_size = 11) +
  labs(
    title = "Empirical Distribution Recovery under Informative Sampling",
    subtitle = "Survey-weighted LBBNN models eliminate selection bias toward the true population",
    x = "Outcome Y",
    y = "Density"
  )
```

Example 2: Two-Pass Differential Privacy Pseudo-Posterior Synthesizer

Simulating Skewed Sensitive Microdata

Following Savitsky, Williams & Hu (2022, Section 5 & Appendix B), financial expenditures and wages are generated from a log-normal distribution, with extreme outliers injected to represent high disclosure risk:

```
set.seed(42)
n_dp <- 300
p_dp <- 4
```

```r
x_dp <- matrix(rnorm(n_dp * p_dp), nrow = n_dp, ncol = p_dp)
colnames(x_dp) <- paste0("x", 1:p_dp)

# Skewed latent process with extreme outliers
latent_mu <- 1.5 + 0.8 * x_dp[, 1] - 0.5 * x_dp[, 2]
y_dp <- exp(latent_mu + rnorm(n_dp, 0, 0.4)) # Log-normal expenditure

# Inject extreme disclosure-risk outliers on records with highest baseline predictions
outlier_idx <- order(latent_mu, decreasing = TRUE)[1:3]
y_dp[outlier_idx] <- y_dp[outlier_idx] * 8.0

# Log-transformed expenditure for Gaussian synthesizer
log_y_dp <- log(y_dp)

cat(sprintf("Dataset size: %d records\n", n_dp))
cat(sprintf("Expenditure scale: Min=%.2f, Median=%.2f, 90th=%.2f, Max=%.2f\n",
            min(y_dp), median(y_dp), quantile(y_dp, 0.90), max(y_dp)))
cat(sprintf("Log scale:         Min=%.2f, Median=%.2f, 90th=%.2f, Max=%.2f\n",
            min(log_y_dp), median(log_y_dp), quantile(log_y_dp, 0.90), max(log_y_dp)))
```

Running the Two-Pass DP Synthesizer (`fit_dp_lbbnn`)

We synthesize on the properly specified log scale $\log(Y)$ using `fit_dp_lbbnn()`: - **Pass 1**: Fits an unweighted model and automatically estimates the residual noise scale $\hat{\sigma} = \mathrm{sd}(y - \hat{y}_{\mathrm{p1}})$ (`sigma = NULL`). - **Monotone Surgical Weights**: Evaluates record sensitivities $f_i = \max_s |L_{s,i}|$ with calibrated $\hat{\sigma}$, computes smooth weights $\alpha_i \in [0, 1]$ ($c_{\mathrm{scale}} = 0.85$, $g_{\mathrm{shift}} = 0.0$), caps weights at M_{thresh}/f_i, and surgically zeroes extreme tail outliers ($f_i > 3M_{\mathrm{thresh}}$). - **Pass 2**: Fits a fresh model weighted by α_i (`normalize_weights = FALSE`). - **Guarantee**: Evaluates the local Lipschitz bound $\Delta_{\alpha,\mathbf{y}} \leq M_{\mathrm{thresh}}$ and local differential privacy parameter $\epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}}$.

```r
spec_dp <- list(
  problem_type = "regression",
  sizes = c(p_dp, 16, 16, 1),
  prior = c(0.5, 0.5, 0.5),
  std = c(1.0, 1.0, 1.0),
  inclusion_inits = "balanced",
  input_skip = FALSE,
  flow = FALSE,
  device = "cpu"
)

# Run Two-Pass Synthesizer with automatic residual scale estimation and monotone truncation
dp_results <- fit_dp_lbbnn(
  model_spec = spec_dp,
  X = x_dp,
  y = log_y_dp,
  epochs = 40,
  lr = 0.02,
  batch_size = 32,
  S = 50,
  c_scale = 0.85,
  g_shift = 0.0,
```

```
  M_thresh = 5.5,
  sigma = NULL, # Automatic residual scale calibration
  device = "cpu",
  verbose = FALSE
)

cat("Two-Pass DP Synthesis Completed!\n")
cat(sprintf("Calibrated Residual Scale (sigma_hat): %.4f\n", dp_results$sigma_hat))
cat(sprintf("Pass 1 Sensitivity Range (f_i):       [%.3f, %.3f] (Median: %.3f)\n",
            min(dp_results$f_vec), max(dp_results$f_vec), median(dp_results$f_vec)))
cat(sprintf("Privacy Weights Range (alpha_i):      [%.3f, %.3f] (Median: %.3f, Zero count: %d)\n",
            min(dp_results$alpha), max(dp_results$alpha), median(dp_results$alpha), sum(dp_results$alpha
cat(sprintf("Local Lipschitz Bound (Delta_alpha):  %.3f\n", dp_results$delta_alpha))
cat(sprintf("Local Privacy Guarantee (Epsilon_y):  %.3f\n", dp_results$epsilon_local))
```

Inspecting Record Sensitivities & Surgical Downweighting

```
risk_df <- data.frame(
  ObsID = seq_len(n_dp),
  Expenditure = y_dp,
  Log_Expenditure = log_y_dp,
  Sensitivity = dp_results$f_vec,
  Alpha = dp_results$alpha,
  IsOutlier = seq_len(n_dp) %in% outlier_idx
)

p1 <- ggplot(risk_df, aes(x = Expenditure, y = Sensitivity, color = Alpha)) +
  geom_point(size = 2.5, alpha = 0.85) +
  scale_color_gradient(low = "#e74c3c", high = "#27ae60", name = expression(alpha[i])) +
  theme_minimal(base_size = 11) +
  labs(
    title = "Sensitivity vs. Expenditure",
    x = "Observed Expenditure ($)",
    y = expression("Record Sensitivity " * f[i])
  )

p2 <- ggplot(risk_df, aes(x = Expenditure, y = Alpha, color = IsOutlier)) +
  geom_point(size = 2.5, alpha = 0.85) +
  scale_color_manual(values = c("FALSE" = "#2980b9", "TRUE" = "#c0392b"), name = "Injected Outlier") +
  theme_minimal(base_size = 11) +
  labs(
    title = "Surgical Privacy Weights",
    x = "Observed Expenditure ($)",
    y = expression("Privacy Weight " * alpha[i])
  )

gridExtra::grid.arrange(p1, p2, ncol = 2)
```

The monotone surgical construction performs cleanly: - Standard records receive privacy weights $\alpha_i \approx 0.65 - 0.85$, allowing them to contribute fully to model training. - As expenditure increases into the moderate tail, weights are smoothly capped at $\alpha_i \leq M_{\text{thresh}}/f_i$. - All three injected extreme outliers ($Y \in [\$400, \$530]$)

produce sensitivities $f_i > 25.0 > 3M_{\text{thresh}}$ and are **surgically truncated to** $\alpha_i = 0$, strictly guaranteeing $\Delta_{\alpha,\mathbf{y}} = 4.32 \leq M_{\text{thresh}}$ and $\epsilon_{\mathbf{y}} = 8.63$.

Comparing Stage 1 & Stage 2 Weight Calibration Schemes (Savitsky, Oct 29, 2026)

To understand how each Stage 1 normalization and Stage 2 reweighting strategy behaves on this sensitive expenditure dataset, we calibrate weights across the four key strategies proposed by Savitsky (2026) using `compute_dp_weights()` at $M_{\text{thresh}} = 5.5$:

```
f_curr <- dp_results$f_vec
M_curr <- 5.5

sch_lin_surg <- compute_dp_weights(f_curr, stage1_method = "linear", stage2_method = "surgical",
                                   c_scale = 0.85, M_thresh = M_curr)
sch_lin_hard <- compute_dp_weights(f_curr, stage1_method = "linear", stage2_method = "hard_trunc",
                                   c_scale = 0.85, M_thresh = M_curr)
sch_pct_hard <- compute_dp_weights(f_curr, stage1_method = "percentile", stage2_method = "hard_trunc",
                                   M_thresh = M_curr)
sch_clip_vec <- compute_dp_weights(f_curr, stage1_method = "target_clip", stage2_method = "vector_upscal
                                   Delta_goal = M_curr, M_thresh = M_curr)

comp_df <- data.frame(
  `Stage 1 Method` = c("linear (c=0.85)", "linear (c=0.85)", "percentile (p=0.05, 0.95)", "target_clip")
  `Stage 2 Method` = c("surgical (Baseline)", "hard_trunc", "hard_trunc", "vector_upscale"),
  `Effective N (sum alpha)` = c(round(sum(sch_lin_surg$alpha), 1), round(sum(sch_lin_hard$alpha), 1),
                                round(sum(sch_pct_hard$alpha), 1), round(sum(sch_clip_vec$alpha), 1)),
  `Full Weight (alpha=1)` = c(sprintf("%.1f%%", 100 * mean(sch_lin_surg$alpha >= 0.999)),
                              sprintf("%.1f%%", 100 * mean(sch_lin_hard$alpha >= 0.999)),
                              sprintf("%.1f%%", 100 * mean(sch_pct_hard$alpha >= 0.999)),
                              sprintf("%.1f%%", 100 * mean(sch_clip_vec$alpha >= 0.999))),
  `Zeroed (alpha=0)` = c(sum(sch_lin_surg$alpha == 0), sum(sch_lin_hard$alpha == 0),
                         sum(sch_pct_hard$alpha == 0), sum(sch_clip_vec$alpha == 0)),
  `Realized Delta` = c(round(sch_lin_surg$delta_alpha, 2), round(sch_lin_hard$delta_alpha, 2),
                       round(sch_pct_hard$delta_alpha, 2), round(sch_clip_vec$delta_alpha, 2)),
  `Local Epsilon` = c(round(sch_lin_surg$epsilon_local, 2), round(sch_lin_hard$epsilon_local, 2),
                      round(sch_pct_hard$epsilon_local, 2), round(sch_clip_vec$epsilon_local, 2)),
  check.names = FALSE
)

knitr::kable(
  comp_df,
  caption = "Comparison of Stage 1 Normalization and Stage 2 Reweighting Strategies (Savitsky, Oct 29, 1
)
```

Key Takeaways from Calibration Comparison: 1. **Target Ratio Clipping with Vector Upward Rescaling (`target_clip` + `vector_upscale`)** achieves the highest Effective Sample Size and retains full weight ($\alpha_i = 1$) on the vast majority of non-outlier units ($f_i \leq M_{\text{thresh}}$), zeroing zero records and smoothly spending the privacy budget via $\min(1, M_{\text{thresh}}/f_i)$. 2. **Percentile Deficit Truncation (`percentile` + `hard_trunc`)** aggressively boosts weights on low-sensitivity units to $\alpha_i = 1$, but hard-truncates the top 5% tail outliers to $\alpha_i = 0$. 3. **Linear Min-Max (`linear`)** scales intermediate units proportionally, and pairing with `surgical` capping prevents sudden threshold discontinuities compared to `hard_trunc`. 4. Crucially, all four schemes guarantee exact sensitivity control $\Delta_{\alpha,\mathbf{y}} \leq M_{\text{thresh}} = 5.50$, proving mathematical $(\epsilon_{\mathbf{y}}, \delta)$-probabilistic differential privacy.

Generating and Benchmarking Synthetic Datasets

Sampling from the **posterior predictive distribution** incorporates parameter uncertainty and residual observation noise:

$$y_i^{\text{synth}} \sim \mathcal{N}\big(\mu_\theta(\mathbf{x}_i), \hat{\sigma}^2\big), \quad Y_i^{\text{synth}} = \exp\big(y_i^{\text{synth}}\big)$$

```
# Non-private baseline synthesizer wrapper
nonprivate_fit <- list(model = dp_results$model_unweighted, sigma_hat = dp_results$sigma_hat)

# Draw 20 synthetic replicate datasets of size n = 300
synth_mat_nonprivate <- sample_synthetic_lbbnn(nonprivate_fit, newdata = x_dp, draws = 20, log_scale = 
synth_mat_private    <- sample_synthetic_lbbnn(dp_results, newdata = x_dp, draws = 20, log_scale = TRUE

calc_synth_stats <- function(mat) {
  rowMeans(apply(mat, 2, function(col) {
    c(mean(col), sd(col), quantile(col, 0.25), median(col),
      quantile(col, 0.75), quantile(col, 0.90), max(col))
  }))
}

clean_y <- y_dp[-outlier_idx]
clean_stats <- c(mean(clean_y), sd(clean_y), quantile(clean_y, 0.25), median(clean_y),
                 quantile(clean_y, 0.75), quantile(clean_y, 0.90), max(clean_y))
orig_stats <- c(mean(y_dp), sd(y_dp), quantile(y_dp, 0.25), median(y_dp),
                quantile(y_dp, 0.75), quantile(y_dp, 0.90), max(y_dp))
nonpriv_stats <- calc_synth_stats(synth_mat_nonprivate)
priv_stats    <- calc_synth_stats(synth_mat_private)

# Distributional Benchmark Table on Expenditure Scale ($)
bench_df <- data.frame(
  Metric = c("Mean", "Std. Dev.", "25th Pct", "Median", "75th Pct", "90th Pct", "Max (Extreme)"),
  Original_Clean = clean_stats,
  Original_With_Outliers = orig_stats,
  NonPrivate_Synthetic = nonpriv_stats,
  DP_Private_Synthetic = priv_stats,
  row.names = NULL
)

knitr::kable(
  bench_df,
  digits = 2,
  row.names = FALSE,
  col.names = c("Metric", "Original (Clean)", "Original (With Outliers)", "Non-Private Synthetic", "DP-
  caption = "Distributional Utility Comparison: Original (Clean vs. Outlier-Contaminated) and Synthetic
)
```

The benchmark table demonstrates the utility and privacy advantages of the two-pass synthesizer: - In the raw data, the 3 extreme outliers (`Max = $529.57`) distort the sample mean (`$11.30` vs. `$6.70` clean) and explode the sample standard deviation (`$46.75` vs. `$7.23` clean). - The `DP-Private Synthetic` data closely matches the **clean population distribution** across the mean (`$7.99`), standard deviation (`$13.95`), median (`$4.13`), and 90th percentile (`$17.10`), while completely eliminating the extreme `$529.57` outlier disclosure risk.

```r
ecdf_df <- data.frame(
  Value = c(y_dp, synth_mat_nonprivate[, 1], synth_mat_private[, 1]),
  Source = factor(c(
    rep("Original Data", n_dp),
    rep("Non-Private Synthetic", n_dp),
    rep("DP-Private Synthetic", n_dp)
  ), levels = c("Original Data", "Non-Private Synthetic", "DP-Private Synthetic"))
)

ggplot(ecdf_df, aes(x = Value, color = Source)) +
  stat_ecdf(geom = "step", size = 1.1) +
  coord_cartesian(xlim = c(0, quantile(y_dp, 0.95))) +
  scale_color_manual(values = c(
    "Original Data" = "#2c3e50",
    "Non-Private Synthetic" = "#e67e22",
    "DP-Private Synthetic" = "#27ae60"
  )) +
  theme_minimal(base_size = 11) +
  labs(
    title = "Synthetic Data Utility: Empirical CDF",
    subtitle = "DP-Private synthetic microdata accurately reproduces the body of the true distribution"
    x = "Expenditure ($)",
    y = "Cumulative Probability"
  )
```

Example 3: Custom Composite Likelihood Integration

The `problem_type = "custom"` functionality allows users to supply custom link and negative log-likelihood (NLL) functions. This serves as the direct architectural bridge to **Track 2** (Vladislav Beresovsky's composite Bernoulli likelihoods for combining probability and non-probability samples).

Specifically, during `train_lbbnn()`, the training loop evaluates `self$nll(input, target, reduction = "none")` when unit weights u_i are present (multiplying each observation's loss contribution ℓ_i by u_i before summation), and `loglik_contributions()` evaluates `reduction = "none"` to populate the $S \times n$ sensitivity matrix $L_{s,i} = -\ell_i(\theta^{(s)})$. This unified interface allows `LBBNN` to seamlessly optimize multi-column targets `cbind(I_z, pi_r)` in Track 2 (`nll_stack`, `nll_link1`, `nll_link2`, `make_nll_mip`, `nll_pseudo`).

```r
# Define custom link: Sigmoid activation
custom_sigmoid_link <- function(x) torch::torch_sigmoid(x)

# Define custom negative log-likelihood with numerical clamping
custom_bce_nll <- function(input, target, reduction = "sum") {
  eps <- 1e-7
  p_hat <- torch::torch_clamp(input, eps, 1.0 - eps)
  t_flt <- target$squeeze()$to(dtype = torch::torch_float32())
  p_flt <- p_hat$squeeze()

  # Pointwise Bernoulli NLL
  nll_i <- -(t_flt * torch::torch_log(p_flt) + (1.0 - t_flt) * torch::torch_log(1.0 - p_flt))
```

```
  if (reduction == "none") {
    return(nll_i)
  } else {
    return(torch::torch_sum(nll_i))
  }
}

# Instantiate custom model
custom_model <- lbbnn_net(
  problem_type = "custom",
  sizes = c(3, 8, 1),
  prior = c(0.5, 0.5),
  std = c(1.0, 1.0),
  inclusion_inits = "balanced",
  link = custom_sigmoid_link,
  nll = custom_bce_nll,
  device = "cpu"
)

# Synthetic binary task
x_cust <- matrix(rnorm(60 * 3), 60, 3)
y_cust <- rbinom(60, 1, 0.5)
dl_cust <- torch::dataloader(
  torch::tensor_dataset(torch::torch_tensor(x_cust), torch::torch_tensor(y_cust)),
  batch_size = 20,
  shuffle = FALSE
)

# Train model
train_res_cust <- train_lbbnn(epochs = 10, LBBNN = custom_model, lr = 0.02, train_dl = dl_cust, verbose

# Extract record-level log-likelihood contributions
L_custom <- loglik_contributions(custom_model, dl_cust, draws = 10)

cat(sprintf("Custom model training loss at epoch 10: %.3f\n", tail(train_res_cust$loss, 1)))
cat(sprintf("Log-Likelihood Contributions Matrix shape: %d draws x %d records\n",
            nrow(L_custom), ncol(L_custom)))
cat(sprintf("All log-likelihood evaluations finite: %s\n", all(is.finite(L_custom))))
```

Collaborator API Reference

All observation-weighted and differential privacy utilities are exported in `LBBNN`:

References

1. **Bhattacharya, A., Pati, D., & Yang, Y.** (2019). *Bayesian fractional posteriors.* The Annals of Statistics, 47(1), 39–66.

2. **Dimitrakakis, C., Nelson, B., Mitrokotsa, A., & Rubinstein, B. I.** (2017). *Differential privacy for Bayesian inference through posterior sampling.* Journal of Machine Learning Research, 18(1), 343–381.
3. **Dwork, C., & Roth, A.** (2014). *The algorithmic foundations of differential privacy.* Foundations and Trends in Theoretical Computer Science, 9(3–4), 211–407.
4. **Savitsky, T. D., & Toth, D.** (2016). *Bayesian estimation under informative sampling.* Electronic Journal of Statistics, 10(1), 1677–1708.
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6. **Wenzel, F., Roth, K., Veeling, B. S., Swiatkowski, J., Tran, L., Mandt, S., Snoek, J., Salimbeni, H., Jenatton, R., & Nowozin, S.** (2020). *How good is the Bayes posterior in deep neural networks really?* Proceedings of the 37th International Conference on Machine Learning (ICML).

Session Information

```
sessionInfo()
```